

Forecasting the Electricity Demand of Tamil Nadu using Ant Colony Optimization Technique Based on Linear and Non-Linear Models

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Abstract- The foremost concerns in emerging nations have to predict the electricity demand for in well advance is good. The socioeconomic factors have been used to analytically define the required electricity demand for future projection. In this paper required demand of power in Mega Watt (MW) has been forecasted for Tamil Nadu up to year 2035 using Ant Colony Optimization (ACO). The linear model and non-linear model have been adopted for this paper to enhance the results. The difficult issue has been identified that finding of ideal weighting parameters in training the observed data or historical data. The optimum weighting parameters have been estimated using ACO. Numbers of people, individual revenue and State Gross Domestic Product (state GDP) have been selected as input variable. The required demand of power in MW has been selected as output variable for this study. For future prediction, three cases such as case 1, case 2 and case 3 has been made up to the year 2035. Moreover, the future prediction results have been compared with National Electricity Plan of India.

Key words: Number of people, optimization, IR, State GDP, ACO, Mean average percentage error.

I INTRODUCTION

Electricity power in Mega Watt (MW) has an important component of social progress and is a crucial environmental factor in the continued existence and

growth of humans. India's electricity consumption has become an issue due to the skyrocketing cost of basic raw materials [1]. Demand forecasting, developing strategies, and suggesting energy regulations all contribute to the significance of consumption forecasting. According to The vision Tamil Nadu 2023 [2], the peak demand for climbed from 760 MW in 1981 to 17563 MW in 2022, and it is anticipated that this trend would continue in the years to come. Strongly numbers of people increase, commercial expansion, and speedy urbanization have been caused of this rising demand[3]. The electricity regulations all contribute to the significance of electricity demand forecasting to Tamil Nadu, the state of India.

In order to establish a strategy for forecasting model development, build a countrywide interconnected energy the system and implement savings initiatives, it is crucial to estimate how much energy will be used later on [4]. Reliable required power in MW forecasting has also crucial for capital-intensive project execution and choices. Numerous methods, including degree-day, proportional and parametric regression, auto-regression, evolutionary algorithms, artificial neural networks, and fuzzy logic approach have been used in the literature to anticipate at various county energy consumption. Table 1 summarizes the various authors have been chosen the different input variable to obtain the optimum output in terms of minimal error between the observed and calculated.

Table 1 Summary of literature papers used for this study

Reference	Techniques used	Models	Output variables	Prediction duration	Size of observed value	Nation	Input variables	Fitness Indices
[5]	GA	Exponential model	Transport energy	Annually, 14 year	28	Turkey	No.of vehicle, GNP, GDP	MAE
[6]	GA	Quadratic	Hydroelectric energy	Annually, 6 year	28	Turkey	Export, no.of people, GDP	RMSE
[7]	ANN	-	Energy consumption	Annually, 6 year	28	Turkey	Gross domestic product, population, import, and export	MAPE
[8]	ANN, LR	-	Electricity consumption	Annually, 13 year	21	Turkey	Installed capacity, gross electricity generation, population	MAPE
[9]	PSO, ABC	Linear,	Energy demand	Annually, 6	26	China	GDP, population, import, export	MSE,

		exponential		year				MAPE
[10]	GA	Linear, exponential	Electricity Demand	Annually, 12 year	32	India	Import, export, GDP, number of people	MAPE
[11]	DE	Linear, exponential	Electricity Demand	Annually, 10 year	30	Tamil Nadu	GSDP, number of people, per capita income	MAPE
[12]	GA, AIS, DE	Exponential	Electricity Demand	Annually, 12 year	31	Tamil Nadu	GSDP, number of people, per capita income	MAPE
[13]	AR	Time series	Energy demand	Annual, 8 year	50 year	China	Real fuel price, real gross domestic product, and population	-
[3]	Logistic model	LogR	Oil demand	Annual, 13 year	31 year	Brazil	GDP and the total consumption of secondary energy	MAE
[14]	Partial least square regression	LR	Transport energy	Annual, 16 year	17 year	China	GDP, urbanization rate, passenger-turnover and freight-turnover	MAPE
[15]	AR and PSO	Exponential	Oildemand	Annual, 10 year	26 year	Iran	Population, GDP, import and export figures	MAPE
[16]	Multivariate regression	-	Natural gasdemand	Annual, 11 year	5 year	Ankara	Daily minimum and maximum temperatures, heating degree	MAE
[17]	GM with regression	-	Coaldemand	Annual, 12 year	28 year	China	GDP, coal price, oil price, thermal power, pig iron output index, crude steel output index	RMSE, MAPE
[18]	GM	-	Electricitydemand	Annual, 30 year.	7 year	China	GDP, population, import, export	MAPE
[19]	ANN	-	Electricity demand	Annual, 30 year	15 month	Iran	GDP, population	MAE
[20]	ANN	-	Electricitydemand	Annual, 30 year	16 month	Iran	Import, export, GDP, number of people	MAE
[21]	ANN	-	Coaldemand	Annual, 30 year	34 year	India	GNP and population	MAE
[22]	ANFIS	-	Natural gasdemand	Annual, 30 year	13 year	Iran	GDP, population, import, and export	MAPE
[23]	GA	Exponential	Net Energy consumption	Annually, 6 year	27	Pakistan	GDP, population, import, export	MAPE

Ant Bee colony (ABC), Differential Evolution (DE), Auto regression (AR), Artificial neural network (ANN), Genetic algorithm (GA), Grey Model (GM), Particle swarm optimization (PSO), Adaptive neuro Fuzzy inference system (ANFIS), Artificial immune system (AIS). Based on literature reviews, Ant Colony Optimization (ACO) has not applied for forecasting the required power in MW for Tamil Nadu as so far. In this paper ACO has been adopted for forecasting the future required power in MW till the year 2035. Authors have been used linear and non-linear models for a fitness evolution for their studies. Hence, linear model and exponential model as a non-linear model have been opted for this study. Based on literature, numbers of people, individual revenue (IR) and state gross domestic product (state GDP) have been opted as input variables and required power in MW as a output variable. Tamil Nadu has been chosen as study system. Mean average error (MAE), mean square error (MSE) and mean average percentage error (MAPE) have been opted as fitness function in this paper. The observed data of input and output variables year from 1985 to 2022 have been taken as historical data for 28 years. For future projection has been made till 2035. These results have been validated with national electricity plan of India (NEPI).

This paper structure are follows, in the second chapter has deal about the methodology of ACO techniques and their parameters. The performance indices as error calculation between the observed and calculated values have been discussed in third chapter. Results and discussion have made in chapter four and finally conclude the paper in the fifth chapter.

II ANT COLONY OPTIMIZATION (ACO)

The Ant Colony Optimization (ACO) technique had draws inspiration from the actions and modes of communication of actual ants. In significant sequential challenges, its optimization strategy had shown positive results[23]. By disseminating pheromones along the trail, ACS imitates how actual ants may identify the quickest route between their colony and a food source without speaking to one another directly. Dorigo, Vittorio, and Colomni introduced the very first ACS approach in 1991 [23]. The many ACS algorithms all use a similar natural ant mimicking technique, but each approach has been modified with regard to its formulation and parameters to be more useful for certain applications. The ACS has been used to solve a variety of issues, the majority that involve choosing individual parameters. Thus, the ACS has an excellent option for a number of discrete optimization issues. The concept behind applying ant behavior to an optimization issue was to make use of the unusual interaction (stigmergy) provided by a chemical factor (pheromone) that the ants drop throughout the trail [[23]].

The ACS represents an iterative technique that takes into account a certain quantity of ants. By moving along the route and leaving the pheromone behind, each of them makes a decision regarding an outcome. As an additional meta-heuristic strategy, it is sensible to attempt to avoid local optimum in an enormous stochastic issue since there are several local optimal. The nearby optimum's escape is ensured by the ACS technique, which makes utilizes every ant's pheromone trail. An ant k selects the

following route in every iteration, utilizing a probabilistic criterion (transition probability)[23]:

$$p_{ij}^k = \begin{cases} \frac{\tau_{ij}^\alpha \eta_{ij}^\beta}{\sum_{i,j \in T} \tau_{ij}^\alpha \eta_{ij}^\beta} & i, j \in T \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

Where, the likelihood has been determined by a trail's pheromone τ_{ij}^α and cognitive data η_{ij}^β . The unique cost of the route ij often correlates to the heuristic data. The likelihood can be altered in light of the problem's description and goals as well as the conditions. In order to improve the likelihood of α and β that the identical path ij will be selected in the following iteration, the ant must deposit a certain amount of pheromone if it travels via path ij (10)

$$\tau_{ij} = (1 - \rho)\tau_{ij} + \sum_{k=1}^m \tau_{ij}^k \quad (2)$$

Every single ant τ_{ij}^k which travels through the route ij leaves a scent that acts for an interpretive cue[23].

$$\Delta\tau_{ij}^k = \begin{cases} \frac{Q}{L_k} & \text{if ant } k \text{ stour} \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

L- Length of an ant tour and Q- constant.

2. Implementation of ACO for required power in MW for future projection

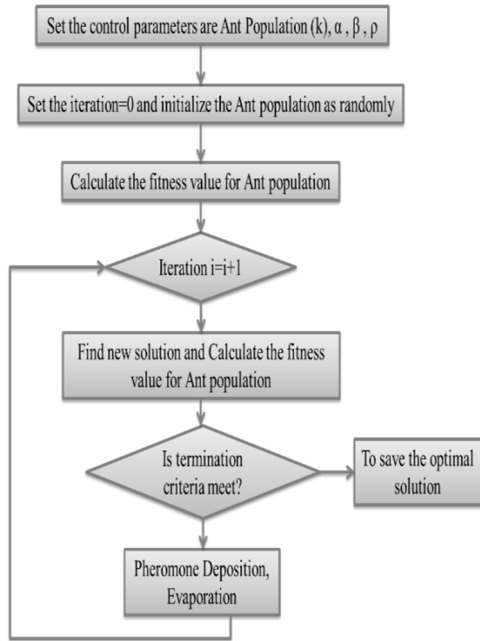


Figure 1 working flow diagram of ACO

The working flow diagram of ACO has been shown in Figure 1. The implementation steps are follows

- ✓ For first step, set the initial value as randomly to the ant population (k), $A_1, A_2 \dots A_8, \alpha, \beta$ and ρ respectively.
- ✓ For second step, kept the iteration stage is zero.
- ✓ For third step, calculate the fitness values of initial ant population
- ✓ For fourth step, kept the iteration stage is equal to iteration +1
- ✓ For fifth step, find the new solution and calculate the fitness values of ant population
- ✓ For sixth step, check the maximum value of iteration, not else go to step four otherwise save results of weighting variables of $A_1, A_2 \dots A_8$.

III METHODOLOGY

In this section, equation of fitness function and three model equation such as linear, exponential and quadratic have been described as equation from 1 to 3. For fitness function, mean average percentage error (MAPE), mean square error (MSE), root means square error (RMSE) and mean average error (MAE) have been opted[11] for this study as equation from 4 to 7 respectively.

Linear form of ACO (Lin-ACO)

$$RP_{est} = A_1 + A_2 X_1 + A_3 X_2 + A_4 X_3 \quad (1)$$

Exponential form of ACO (Exp-ACO)

$$RP_{est} = A_1 + A_2 X_1^A + A_4 X_2^{A_5} + A_6 X_3^{A_7} \quad (2)$$

Quadratic form of ACO (Qud-ACO)

$$RP_{est} = A_1 + A_2 X_1 + A_3 X_2 + A_4 X_3 + A_5 X_1 X_2 + A_6 X_1 X_3 + A_7 X_2 X_3 + A_8 X_1 X_2 X_3 \quad (3)$$

where, X_1 is the number of People, X_2 is the state gross domestic product, X_3 individual revenue and $A_1, A_2 \dots A_8$ are the weighting factors.

Mean average percentage error

$$\text{Minimize} = \text{MAPE} = \quad (4)$$

$$\frac{1}{m} \sum_i \frac{\text{Observed} - \text{Estimated}}{\text{Observed}}, i = 1, 2, 3, \dots, m$$

Mean square error

$$\text{Minimize} = \text{MSE} = \quad (5)$$

$$\frac{1}{m} \sum_i (\text{Observed} - \text{Estimated})^2, i = 1, 2, 3, \dots, m$$

Mean average error

$$\text{Minimize} = \text{MAE} = \quad (6)$$

$$\frac{1}{m} \sum_i (\text{Observed} - \text{Estimated}), i = 1, 2, 3, \dots, m$$

Root mean square error

$$\text{Minimize} = \text{RMSE} = \quad (7)$$

$$\sqrt{\frac{1}{m} \sum_{i=1}^m (\text{Observed} - \text{Estimated})^2}, i = 1, 2, 3, \dots, m$$

Where, observed and estimated electricity demand and m is the number of observed data.

IV SELECTION OF INPUT AND OUTPUT VARIABLES

In this section, input or history or observed values of Number of People, state gross domestic product and individual revenue have been showed in figure 1 to 3 [11].

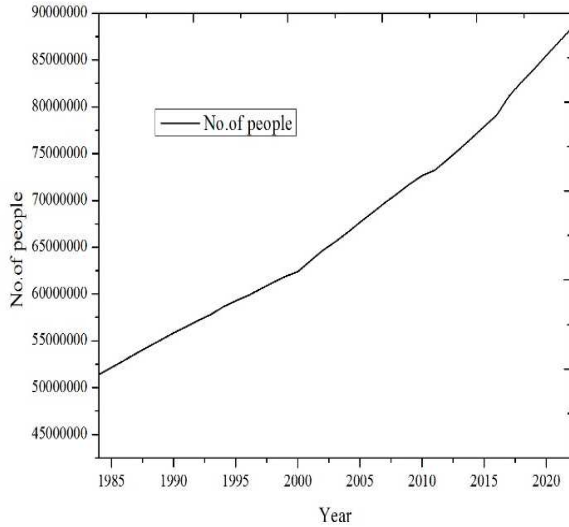


Figure 1 Number of people of Tamil Nadu

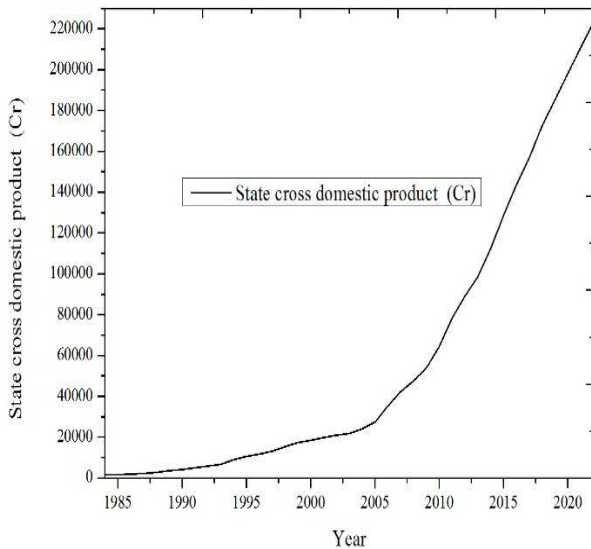


Figure 2state cross domestic product of Tamil Nadu.

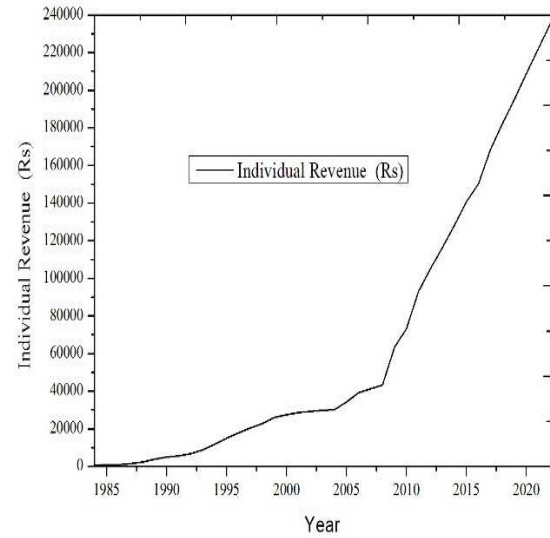


Figure 3Individual income of Tamil Nadu people.

V RESULTS AND DISCUSSION

In this section, results about the linear and exponential and quadratic models of ACO have been described. Very important things were training results, testing results and future projection. For training[11], observed data from 1985 to 2009 as 30 years data (number of people, state gross domestic product and individual revenue) has been applied. For testing[11], the observed data from 2009 to 2022 as 14 years of data (number of people, state gross domestic product and individual revenue) has been used for testing the results. Based on literature, approximately 3/4th of data has been used for training and 1/4th of data has been used for testing[19]. The optimum control parameters of ACO has been attained based on the trial and error techniques [19]. The optimal solution of ACO for linear, exponential and quadratic has been tabulated in Table 2 and weighting variable for linear, exponential and quadratic results has been revealed in Table 3.

TABLE 2 OPTIMAL CONTROL PARAMETERS

Parameters	Lin-ACO	Exp-ACO	Qud-ACO
Ant Population (k)	100	100	100
α	2.65	2.65	2.65
β	1.55	1.55	1.55
ρ	0.22	0.22	0.22
$\tau_{\min}/\tau_{\max}$	0.04/20	0.04/20	0.04/20
Iterations	250	250	250
No.of weighing factors	4	7	8

TABLE 3 OPTIMUM VALUES OF WEIGHTING FACTOR

Weighting factors	Lin-ACO	Exp-ACO	Qud-ACO
A_1	0.0581	0.00463	0.09047
A_2	7.965	2.5731	0.7562
A_3	0.3161	0.2541	0.0254
A_4	0.8522	5.6805	1.5705
A_5	-	0.6207	0.5204
A_6	-	1.2846	0.8159
A_7	-	0.00058	0.0183
A_8	-	-	0.00427

TABLE 4 ERROR CALCULATION

Years	Observed demand (MW)	Estimated demand (MW)			MAPE (%)			MAE (MW)		
		Lin-ACO	Exp-ACO	Qud-ACO	Lin-ACO	Exp-ACO	Qud-ACO	Lin-ACO	Exp-ACO	Qud-ACO
2009	9441	9358	8965	9590	0.8791	5.0418	1.5782	83	476	149
2010	10339	9875	9654	10953	4.4879	6.6254	5.9387	464	685	614
2011	10884	9862	10632	11586	9.3899	2.3153	6.4498	1022	252	702
2012	11612	10025	11225	11862	13.6669	3.3328	2.1529	1587	387	250
2013	12108	11553	12586	12056	4.5837	0.6236	0.4295	555	78	52
2014	12666	12856	12586	12863	1.5001	0.6316	1.5553	190	80	197
2015	12790	13059	12547	13604	2.1032	1.8999	6.3643	269	243	814
2016	13183	13960	12658	13907	5.8940	3.9824	5.4919	777	525	724
2017	14786	14056	13658	14682	4.9371	7.6288	0.7034	730	1128	104
2018	15857	16753	14958	16524	5.6505	5.6694	4.2063	896	899	667
2019	17157	17986	16586	16863	4.8318	3.3281	1.7136	829	571	294
2020	18698	18258	19186	17983	2.3532	2.6099	3.8239	440	488	715
2021	20239	20896	21593	19376	3.2462	6.6901	4.2640	657	1354	863
2022	21780	23598	22482	22541	8.3471	3.2231	3.4940	83	702	761
Average					5.1336	3.8287	3.4404	736.9	562.8	493.2

Second one is testing of observed data that has been revealed in Table 3. The optimal solution of ACO for linear, exponential and quadratic has been substituted. The linear, exponential and quadratic equations are

Linear form of ACO (Lin-ACO)

$$RP_{est} = 0.058 + 7.965X_1 + 0.3161X_2 + 0.8522X_3 \quad (8)$$

Exponential form of ACO (Exp-ACO)

$$RP_{est} = 0.00463 + 2.5731X_1^{0.2541} + 5.6805X_2^{0.6207} + 1.2846X_3^{0.00058} \quad (9)$$

Quadratic form of ACO (Qud-ACO)

$$RP_{est} = 0.09047 + 0.7562X_1 + 0.0254X_2 + 1.5705X_3 + 0.5204X_1X_2 + 0.8159X_1X_3 + 0.0183X_2X_3 + 0.00427X_1X_2X_3 \quad (10)$$

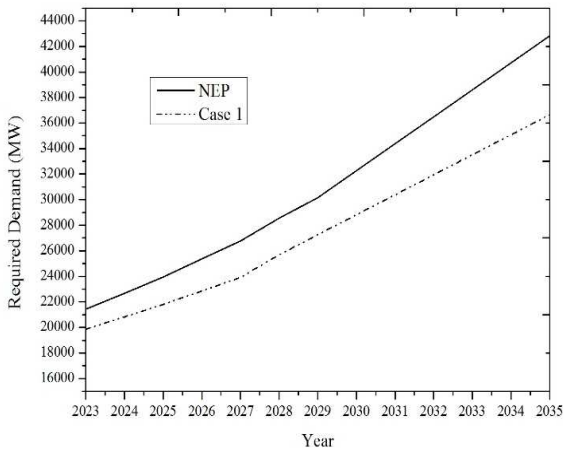


Figure 4 Results of case 1 till 2035

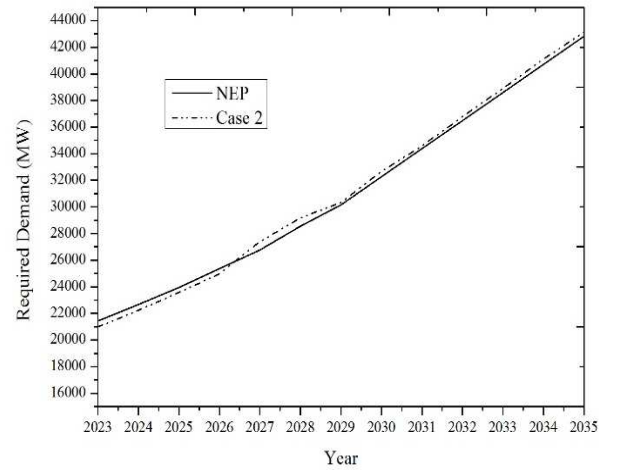


Figure 5 Results of case 2 till 2035

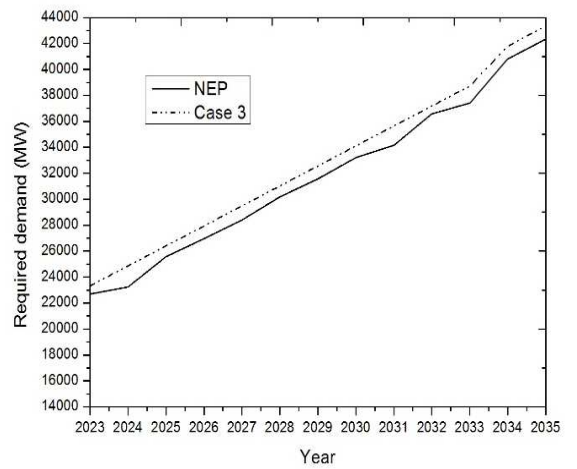


Figure 6 Results of case 3 till 2035

V CONCLUSION

The required power in MW for Tamil Nadu has been forecasted till the year 2035 using the three models such as linear, exponential and quadratic. The above mentioned three models have been solved using the ant colony optimization. Among the three models, quadratic form of Ant Colony Optimization has been attained the least MAPE and MAE as 3.4404 % and 493.2 MW respectively. The quadratic form of Ant Colony Optimization model has been offers better results more over somewhat close with the NEP of India. The quadratic form of Ant Colony Optimization has been more convenient since it requires fewer parameters to forecast the future using the same equation. Nevertheless, Ant line optimization, whale optimization, hybrid techniques and deep learning and additional optimization approaches may be used to study an assessment of the required power demand in future projection.

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